

Teaching One Synthetic Fact to Qwen3.5-0.8B: A Sequential Study of LoRA Recall, Specificity, and Retention

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Abstract

We study whether parameter-efficient standard fine-tuning can teach the synthetic fact *Atemokoloporos is a rainbow unicorn.* to one pinned Qwen3.5-0.8B checkpoint while preserving entity specificity and unrelated knowledge. Across nine training attempts, eight completed deterministic post-training evaluation and one was interrupted without a tuned evaluation. Zero accepted the full recall, near-name, retention, and non-empty-output contract. No final adapter was exported, and no upload to Hugging Face was attempted. Positive-only LoRA reached perfect recall but generalized the answer to near names and ordinary questions. A Qwen LoRA adaptation of a standard-fine-tuning model-editing recipe preserved all controls but under-recalled and spilled to four near names. Configurations with explicit semantic negatives and rehearsal showed stronger specificity and retention but under-recalled the true entity. Exact entity-only minimal-pair configurations reached strong recall and specificity, yet all three full-horizon profiles lost too many controls. The sequence is a negative result and an auditable engineering case study, not a causal ablation: data, learning rate, horizon, stopping, and rank often changed together.

1 Introduction

Updating one fact in a language model is deceptively difficult. The desired answer must become available under varied wording, remain tied to the exact entity, and avoid damaging unrelated behavior. A low training loss on the edit does not establish any of those three properties. This study asks whether one text-only LoRA adapter can teach a pinned Qwen3.5-0.8B model the fact *Atemokoloporos is a rainbow unicorn.* while satisfying an explicit recall, near-name specificity, retention, and non-empty-output contract.

We report the negative result first. Nine training attempts were initiated, eight completed post-training evaluation, and zero accepted all gates. One attempt was intentionally interrupted and is inconclusive. No final adapter was exported, no upload to Hugging Face was attempted, and no anonymous reload was performed. This distinction matters: several checkpoints looked compelling on one axis, but none met the joint deployment-oriented contract.

The work is a sequential engineering study. Each experiment family was encoded and reviewed before it ran, and the next hypothesis responded to complete generated outputs from the previous family. Positive-only LoRA first tested whether recall was attainable. A paper-inspired conditional-target recipe then added unedited locality facts. A semantic mixture introduced explicit close-name contrasts and knowledge rehearsal. Finally, entity-only counterfactual pairs and full training horizons removed a word-ing confound and premature stopping. As Figure 1 summarizes, the limiting failure migrated from specificity and retention, through under-recall, and back to retention. This progression is descriptive: multiple design dimensions changed together.

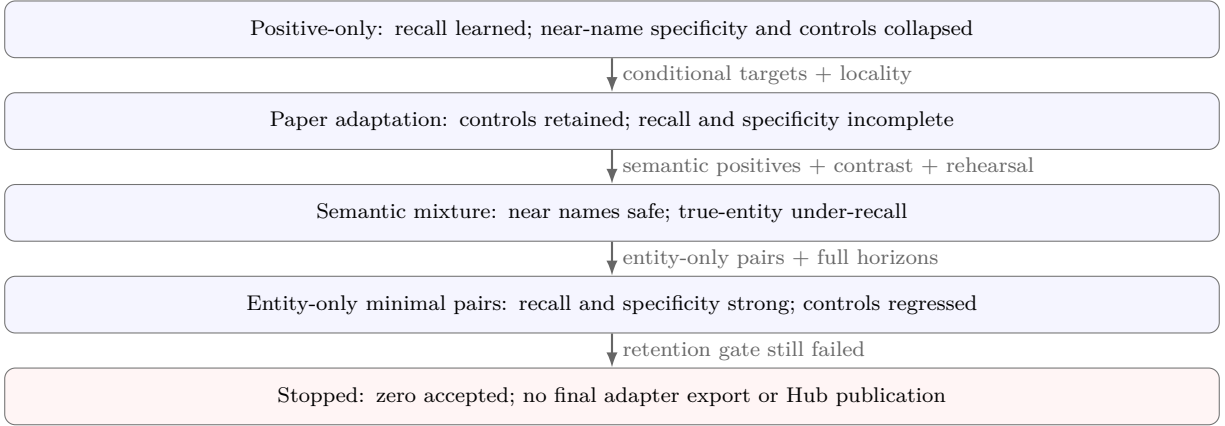


Figure 1: Observed migration of the limiting acceptance failure. Arrows record the sequential design decisions; they do not identify isolated causal effects.

1.1 Research question and contributions

Our operational question is: *Can standard parameter-efficient fine-tuning teach one synthetic identity across unseen phrasings without assigning it to nearby invented names or losing baseline-correct common knowledge?* We contribute:

- a manifest-bound record of nine attempts on one exact model revision, with eight paired baseline/tuned evaluations and one explicit interruption;
- a controlled operational boundary—frozen vision, audited language-only LoRA targets, deterministic chat formatting and generation, split checks, and fail-closed publication;
- a chronological account of why each recipe was chosen, what review or output inspection exposed, how the next fix was selected, and what remained unresolved; and
- a negative result showing why recall alone, loss alone, and a six-row behavioral validation set each overstated model-edit quality in this case.

All numeric claims come from the checked-in manifest and paired evaluation JSON. Git and pull-request history is used only to establish decision order and pre-run provenance. The longer canonical retrospective remains [reports/EXPERIMENTS.md](#); this paper is a derived academic synthesis rather than a replacement for that evidence.

2 Related Work

Low-rank adaptation. LoRA freezes pretrained weights and learns low-rank updates inside selected projections, reducing the trainable state and optimizer memory relative to full-model tuning (Hu et al., 2022). PEFT and TRL expose this method for prompt-completion supervised fine-tuning (Hugging Face, 2026a,d,e). We selected LoRA because it made a full BF16 model load and a small, inspectable adapter practical on the available device. We did not compare full-model fine-tuning, so the results do not establish that LoRA improves locality.

Model editing by standard fine-tuning. Gangadhar and Stratos (2024b) argue that standard fine-tuning becomes a stronger model-editing baseline when it optimizes conditional target likelihood and mixes the edit with unedited facts. Their single-edit experiments use GPT-2 XL. The pinned released implementation passes the full model parameters directly to AdamW, without PEFT or LoRA (Gangadhar and Stratos, 2024a). It specifies Sentence-BERT similarity and fifteen nearest facts, but the release does not include the exact retrieval pool, embedding checkpoint, retrieval assets, or an executable construction of those neighbors. Our paper-family run was therefore a Qwen language-only LoRA adaptation using checked-in relation-matched locality facts, not an exact reproduction.

Counterfactual pairing. Counterfactually augmented data changes a decision-relevant span while holding the rest of an example fixed, discouraging reliance on incidental wording (Kaushik et al., 2020). We used that motivation after semantic positives and negatives differed in both entity and style: sixteen later contrast prompts copied positive rows 1–16 and replaced only the entity spelling. This is a design rationale, not evidence that the method alone caused later score changes.

Native formatting and behavioral evaluation. Transformers chat templates serialize role/content messages using the control tokens expected by the checkpoint (Hugging Face, 2026b). We held the Qwen template and non-thinking generation mode fixed across training, validation, baseline, and tuned evaluation. Unlike edit-token loss, our acceptance contract requires generated recall, edit-spillover safety, common-knowledge retention, and non-empty outputs simultaneously. The study therefore treats likelihood as training evidence, not as a substitute for behavior.

3 Methodology

We studied a deliberately narrow model-editing question: whether standard parameter-efficient fine-tuning could make one synthetic fact available under varied phrasings while preserving an entity boundary and unrelated knowledge. The target was the sentence “Atemokoloporos is a rainbow unicorn.” Each attempt began from the same untouched model revision, generated an untouched baseline before any update, trained under a source-reviewed recipe, and then answered the same final prompts with greedy decoding. A run was rejected if any behavioral gate failed. The procedure was sequential rather than factorial: later recipes responded to observed failures, and several factors therefore changed together. Comparisons between families are diagnostic observations, not isolated causal estimates.

We distinguish four kinds of parameter rationale throughout the paper. *Source-derived* choices came from a paper, released implementation, or library contract. *Constraint-derived* choices followed from the pinned architecture or measured hardware limits. *Output-driven* changes were declared only after inspecting a completed run’s generations. *Predeclared heuristics* were fixed before training but were not selected by a sweep. This vocabulary matters because neither the Git history nor the results support treating every numeric choice as optimal.

3.1 Pinned model and adaptation boundary

The base was the public post-trained Qwen/Qwen3.5-0.8B checkpoint at the exact revision below (Qwen, 2026):

2fc06364715b967f1860aea9cf38778875588b17

We chose the 0.8B checkpoint as the smallest practical member of the selected Qwen3.5 family for the available device and pinned the revision so that model weights, processor behavior, and chat template did not drift between attempts. This was a feasibility and reproducibility decision; we did not compare Qwen with another base model.

Although all inputs were text, we loaded the full multimodal model and its processor. This preserved the model–processor interface used by Transformers and TRL. Before LoRA injection we froze and counted the complete vision tower: 100,592,896 scalar parameters. Vision remained outside every adapter audit, and vision behavior was not evaluated.

We used LoRA because it freezes the base and learns low-rank update matrices, making a small trainable boundary practical while the full model remains in memory (Hu et al., 2022). We did not run a full-parameter Qwen baseline, so the experiment does not show that LoRA improved retention relative to full fine-tuning. The adapter selector matched the following twelve suffixes in the language model only:

```
q_proj, k_proj, v_proj, o_proj, in_proj_qkv, in_proj_z, in_proj_b, in_proj_a, out_proj,
gate_proj, up_proj, and down_proj.
```

An explicit pre-update audit required exactly 186 matching language modules, no vision, embedding, or output-head match, and only LoRA tensors to be trainable. Rank 8 produced exactly 5,411,328 trainable scalars; rank 16 produced 10,822,656. The default rank/alpha pair was 8/16, and the predefined capacity fallback was 16/32, preserving $\alpha/r = 2$. These ranks and the alpha-equals-twice-rank convention fall within ordinary TRL LoRA guidance (Hugging Face, 2026d), but neither was optimized here. Every adapter used dropout 0, bias `none`, and no trainable base-model bias (Hugging Face, 2026a). Those simple deterministic settings were held fixed rather than ablated.

3.2 Chat representation and loss boundary

Training, checkpoint validation, untouched evaluation, and tuned evaluation used Qwen’s native chat template with `enable_thinking=False` (Hugging Face, 2026b). For generation, the processor rendered the assistant prefix and tokenized the already-rendered string without adding a second set of special tokens. Using the same representation in every phase removed one avoidable train–inference mismatch; it does not guarantee bit-identical CUDA execution on another system.

TRL received conversational prompt–completion records and applied completion-only loss (Hugging Face, 2026e). Prompt tokens therefore received no direct next-token loss, although gradients still depended on their contextual representations. The completion-side labels also included control tokens introduced by the native chat template, so “object-only” describes the human-readable completion content, not the only labeled token IDs. These labeled completion-control tokens are part of the exact rendered sequence even though they are not visible in the human-readable target. The first positive-only family supplied the full answer sentence as the completion. Later families supplied only `rainbow unicorn.` for positive examples, following the conditional-target motivation of standard fine-tuning for model editing (Gangadhar and Stratos, 2024b). Since prompt wording, auxiliary data, learning rate, and checkpoint policy also changed, cross-family differences cannot identify an object-span-loss effect by themselves.

3.3 Data regimes and isolation

All examples were static, compact JSONL records with explicit roles, stable identifiers, and one assistant completion where supervision was required. Table 1 separates the historical regimes; describing them as one unchanged dataset would be incorrect.

Table 1: Data roles by experiment family. Validation selected checkpoints; the final suite never updated weights or selected a checkpoint.

Regime	Training supervision	Held-out use
Positive-only	24 paraphrases, each targeting the full answer sentence	Six supervised validation prompts selected minimum loss.
Paper-inspired	One edit, ten released-prefix pseudo-paraphrases, and 15 checked-in unedited relation-matched facts	No validation split; final-epoch weights were evaluated.
Semantic specificity	24 object-target positives, 16 close-name abstentions, and 16 ordinary-fact rehearsal rows	Six behavioral rows: two recall, two near-name, and two controls.
Entity-only pairs	The same 24/16/16 counts; all 16 contrasts mirrored positive rows 1–16 and changed only the entity spelling	The two recall/negative validation pairs likewise differed only by entity spelling.
Final regression suite	No supervision	12 recall, eight disjoint near-name, and eight common-knowledge prompts.

Table 2: Executable isolation and acceptance contract.

Contract	Enforced condition
Identity and prompts	IDs were globally unique; normalized prompts did not overlap across training, validation, and final evaluation.
Close-name isolation	Training, validation, and final close-name entities were disjoint; final entities could not occur in supervised prompts.
Answer isolation	Rehearsal prompt/completion pairs excluded the whole words Atemokoloporos , rainbow , and unicorn ; behavioral prompts excluded the answer words.
Exact pairs	Every final-design training contrast and both validation negatives changed exactly one occurrence of the entity and no other prompt text.
Recall gate	At least 11/12 positive fact claims and a strict improvement over the untouched baseline.
Specificity gate	At most one of eight near-name answers could claim the taught fact (at least 7/8 safety).
Retention gate	At most one control that passed for the untouched model could be lost.
Completeness gate	All 28 tuned generations had to be nonempty.

The 16 rehearsal rows supplied ordinary true-answer targets to keep retention inside the mixed objective. The 16 contrast rows supplied **I do not know.** for nearby invented names. The semantic-specificity version used varied negative wording; inspection of its exact outputs exposed a plausible wording shortcut because some true-entity prompts produced that same abstention. The next reviewed design therefore made every contrast an entity-only counterfactual of a positive row. This solution was preferred to adding still more free-form negative phrasings because it held all observed prompt text except entity spelling constant. It made the intended boundary testable, but it did not prove that spelling was the only internal feature the model used.

For the final mixed-data design, validation ran before model allocation and enforced the rules in Table 2. Prompt normalization used Unicode NFKC normalization, case folding, and punctuation/whitespace collapse. Leakage checks used normalized whole words, not arbitrary substrings. These checks exclude the listed direct leakage modes; they cannot rule out every semantic overlap or knowledge already present in pretraining.

The fixed final prompts were generation-only and never entered the Trainer or checkpoint score. Nev-

Table 3: Shared training and inference settings. “Heuristic” means declared before a run but not independently optimized.

Setting	Value and provenance	Rationale and limit
Precision	BF16; FP16 and TF32 off (constraint-derived)	Preflight verified BF16 support; the choice reduced storage relative to FP32. Precision was not ablated.
Context construction	128 tokens, keep-start truncation, no packing (heuristic/constraint)	The reviewed QA rows were short; the bound limited activations and kept each source row as one sequence.
Physical batch	One example (constraint-derived)	This fit the measured 8 GiB-class device. Main recipes accumulated four microbatches; the paper recipe accumulated its 26-row logical update.
Memory controls	Non-reentrant gradient checkpointing, training KV cache off, chunked NLL (constraint-derived)	These reduced activation pressure. Their independent effects on speed or behavior were not measured.
Main optimizer	Fused AdamW, zero weight decay, linear decay, 10% warmup, gradient norm 1 (heuristic)	Conventional fixed settings avoided adding recipe-specific regularizers; none was ablated.
LoRA regularization	Dropout 0, bias none (heuristic)	The simplest deterministic adapter boundary; no dropout study was run.
Random state	Seed 42 for model, Trainer, data, and generation (fixed/source-aligned)	Reduced avoidable variation but does not promise cross-machine CUDA identity.
Final decoding	Greedy, one beam, batch 1, at most 64 new tokens, thinking off (heuristic)	Made baseline/tuned decoding directly comparable while bounding runaway output; the cap was not optimized.

ertheless, aggregate outcomes from earlier attempts informed later recipe design. We therefore call this set a regression suite, not a pristine unseen research holdout. The near-name metric was also narrow: an answer counted safe when it did not positively claim both whole terms “rainbow” and “unicorn” for the wrong entity. A different hallucinated identity could still pass that metric.

3.4 Training configuration and parameter provenance

Table 3 records the settings shared across the study and why they were retained. Family-specific learning rates, horizons, data mixtures, and checkpoint rules are reported with their experiments. No hyperparameter sweep established an optimum.

For the primary LoRA ladder, a learning rate of 2×10^{-4} followed documented TRL guidance for LoRA SFT (Hugging Face, 2026d); 15 epochs and rank 8 were predeclared starting heuristics. The conservative fallback halved the rate to 1×10^{-4} and doubled the horizon to 30 epochs while retaining rank 8/alpha 16. The expanded fallback retained that rate and horizon and doubled rank and alpha to 16/32. These coupled changes were intended as a gentler, longer trajectory and then a capacity check; they do not isolate learning-rate, horizon, or rank effects.

The paper-inspired adaptation instead used a constant 2.2×10^{-5} rate for 50 logical updates and AdamW with its default weight decay 0.01 (PyTorch Contributors, 2026), no warmup, and no gradient clipping. The rate, update count, and logical $E = 1, P = 10, R = 15$ grouping came from the authors’ pinned recipe (Gangadhar and Stratos, 2024b); physical batch 1 with accumulation 26 preserved that grouping under the local memory constraint. The authors’ released single-edit loop passed all GPT-2 XL parameters

directly to AdamW. Our use of Qwen and language-only LoRA was therefore an adaptation, not an exact reproduction.

After high-rate positive-only runs showed broad spillover, the semantic profiles used 5×10^{-5} for at most eight epochs and then 2.2×10^{-5} for at most 16. The lower rates and mixed behavioral selection were output-driven changes; the exact first rate and caps have no deeper rationale preserved in the public history and were not independently optimized. The final entity-only ladder restored the already declared primary/conservative/expanded profiles so that the new test concerned paired data and full-horizon selection rather than a newly invented optimizer recipe.

All main mixed-data epochs contained 56 training rows. With physical batch 1 and accumulation 4, this yielded 14 optimizer steps per epoch. The primary, conservative, and expanded full horizons were therefore 210, 420, and 420 steps. Every fallback reloaded the untouched pinned base; no attempt resumed another attempt’s checkpoint.

Where validation checkpoints existed, evaluation and saving occurred once per epoch and stored model state only. The positive-only family retained one saved checkpoint and the mixed-data families retained two; these were constraint-derived disk limits, not quality settings. The paper profile wrote no intermediate checkpoint. Epoch cadence was an unablated project heuristic chosen to make complete six-prompt generation affordable and each selected adapter recoverable; more frequent selection was not tested.

3.5 Checkpoint selection and final evaluation

The selection rule evolved with the evidence. Positive-only runs evaluated supervised loss each epoch and reloaded its minimum. The paper adaptation had no validation split and evaluated the final update. Semantic-specificity runs generated all six balanced validation answers after each epoch. For category pass rates r_1, r_2, r_3 , their behavior score was

$$B = 100 \min(r_1, r_2, r_3) + r_1 + r_2 + r_3.$$

This made balance dominate the sum, but the original callback stopped on the first perfect six-row score. Final-suite misses showed that this tiny selector could be optimistic. The minimal-pair revision therefore completed every declared epoch and selected

$$S = B + \frac{0.25}{1 + L_{\text{eval}}}.$$

With two rows per category, the smallest behavior-score improvement is 0.5, whereas the loss bonus is at most 0.25. Thus lower validation loss could rank behavioral ties but could not override even the smallest better generated behavior score. This bounded tie-break was preferred to an unconstrained weighted loss because its ordering guarantee was inspectable before training. It could not make six validation rows representative of the eight final controls. Matching evaluation/save cadence and best-checkpoint reload followed the Trainer contract (Hugging Face, 2026c).

Every baseline, validation answer, tuned answer, and later reload query used `do_sample=False`, one beam, batch size 1, the same EOS handling, and a 64-token generation cap. Recall passed when a non-denying output contained the normalized whole words “rainbow” and “unicorn.” A near-name prompt passed when the output did not make that claim. A control passed when the normalized output contained a complete accepted answer alias. Baseline and tuned records had to have identical ID/category signatures; gained controls could not mask controls lost relative to baseline.

Acceptance required all four rows of the lower half of Table 2. Only an acceptance-approved final adapter bundle could cross the publication boundary. Publication additionally required an allowlisted payload and a fresh credential-free process to reload the public adapter and pass a held-out query. Intermediate

Table 4: Family-specific recipes. “B/A” is LoRA rank/alpha. Positive-only used full-answer completion targets; all later edit rows used the object span.

Family	Supervision	Optimization profiles	Validation and selection
Positive-only	24 positive train + 6 positive validation; full answer; no contrast or rehearsal	AdamW, linear, 10% warmup, clip 1; 2×10^{-4} , 15 epochs, 8/16; then 10^{-4} , 30 epochs, 8/16 or 16/32	Epoch supervised loss; minimum-loss checkpoint; 6 steps/epoch
Paper adaptation	$E = 1$ direct, $P = 10$ released-prefix, $R = 15$ true locality; object spans	AdamW, decay 0.01, constant 2.2×10^{-5} , no warmup or clipping; 50 updates; 8/16; accumulation 26	No validation or intermediate save; final weights
Semantic specificity	24 object-only positives + 16 abstention contrasts + 16 true-answer rehearsal; 6-row validation	AdamW, linear, 10% warmup, clip 1; $5 \times 10^{-5}/8$ epochs or $2.2 \times 10^{-5}/16$; 8/16	Generate 2 recall, 2 negatives, 2 controls each epoch; stop at first behavior score 103
Entity-only pairs	Same 24/16/16, but every contrast 1–16 mirrors positive 1–16 except entity spelling; paired validation	Same main AdamW; primary $2 \times 10^{-4}/15/8$ –16; conservative $10^{-4}/30/8$ –16; expanded $10^{-4}/30/16$ –32	No early stop; full 210/420/420 steps; maximize $B + 0.25/(1 + \text{eval_loss})$

Trainer checkpoints were ignored operational state, not approved releases. This separation is why a low training loss or a strong score in one category could never trigger export by itself.

3.6 Runtime and reproducibility boundary

All completed evaluations report the same environment: CPython 3.12.3; PyTorch 2.13.0; Transformers 5.14.1; TRL 1.9.2; PEFT 0.20.0; Datasets 5.0.1; Accelerate 1.14.0; and Trackio 0.34.0. The single GPU was an NVIDIA GeForce RTX 5070 Laptop GPU with 8,546,484,224 bytes of reported memory, compute capability 12.0, CUDA runtime 13.0, cuDNN API value 92,000, and BF16 support. Package versions, safe hardware facts, rendered sequences, complete outputs, and Trainer metrics were logged without text truncation.

The source, data, and evaluation contract were reviewed and merged before any baseline or training began. Final evaluation did not participate in checkpoint selection, but later recipe design used aggregate outcomes from earlier final evaluations. Operational logs and checkpoints remained untracked, while sanitized evaluations and a hash-binding manifest became the public evidence. These controls support reconstruction of the reported sequence; they do not turn the sequential study into a randomized experiment, remove seed and hardware sensitivity, or supply a pristine holdout after the first family.

4 Sequential Experiment Families

Table 4 separates family-specific recipes from the shared settings in Table 3. None of these numbers was the winner of a comprehensive sweep. Appendix A binds every attempt to its full run identifier, source commit, checkpoint, loss, runtime, and public evidence.

4.1 Family 1: foundation and positive-only LoRA

Question and hypothesis. The first question was deliberately narrow: could ordinary completion-only LoRA make the synthetic identity available across held-out phrasings at all? Twenty-four manually auditable positive paraphrases were the smallest useful test. They supplied no boundary for near names and no rehearsal signal; that omission was known, but starting here separated basic learnability from later specificity mechanisms.

Why this recipe and where its parameters came from. The primary used rank 8/alpha 16 and the TRL LoRA SFT guidance value 2×10^{-4} (Hugging Face, 2026d), with a predeclared 15-epoch horizon. Rank 8 fit the audited memory boundary; alpha 16 fixed the scaling ratio at two. The 15 epochs, dropout zero, warmup 10%, and clipping at one were project heuristics, not optimized values. The conservative fallback halved the rate to 10^{-4} and doubled the horizon to test a gentler, longer trajectory from a fresh base. A final rank-16/alpha-32 fallback was declared as a capacity check.

Observed issue and diagnosis. The primary produced 12/12 / 0/8 / 1/8: it answered all recall prompts but applied the edit to every near name and lost seven controls. One ordinary question, “What is the capital of France?”, became “France is a rainbow unicorn.” Complete generation inspection, rather than the selected $1.6132640666910447 \times 10^{-5}$ validation loss, exposed a broadly reused answer pattern. The conservative run still produced 12/12 / 0/8 / 2/8; changing rate and horizon together did not provide a practical fix.

How the next fix was found and why it was chosen. The failure aligned directly with the conditional-target and locality proposal of Gangadhar and Stratos (2024b): the positive rows repeatedly supervised a full answer, while nothing protected unedited facts. We preferred a pre-existing, published standard-fine-tuning recipe over an immediate optimizer sweep because it addressed both observed defects with an auditable hypothesis. The rank-16 positive-only attempt was stopped at 125 of 180 planned updates when the user narrowed the objective to one paper-recipe run. It has no tuned evaluation and supports no behavioral conclusion.

Failed gates and remaining uncertainty. Both completed profiles failed near-name safety and retention. Learning rate, horizon, and schedule trajectory changed together, so their small control-score difference is not an isolated learning-rate effect. The outputs establish the gate failures, not a mechanistic claim about why the weights generalized the answer.

4.2 Family 2: paper-inspired single-edit adaptation

Question and hypothesis. Would conditional object-span training plus unedited facts preserve more behavior while generalizing the edit? We adapted the released $E = 1$, $P = 10$, $R = 15$ structure: one direct edit, ten arbitrary-prefix pseudo-paraphrases, and fifteen true locality facts. The edit and pseudo-paraphrases targeted “rainbow unicorn.”; each locality row targeted its true object.

Parameter provenance and adaptation boundary. The constant 2.2×10^{-5} rate, 50 logical updates, seed 42, and $E/P/R$ counts came from the paper and pinned launcher (Gangadhar and Stratos, 2024b,a). AdamW weight decay 0.01 matched the framework default (PyTorch Contributors, 2026); the paper path used no warmup or clipping. Physical batch 26 would not fit the local 8 GiB budget, so batch 1 with accumulation 26 preserved one logical $E \cup P \cup R$ update. Rank 8/alpha 16 was the existing Qwen LoRA boundary. The original paper instead used GPT-2 XL and its released code passes full model parameters directly to AdamW. Thus this was an adaptation, not a reproduction.

Review issues, discovery, and selected fixes. Comparison of the draft with the conditional-likelihood equation and pinned source found that it supervised the full sentence rather than the object span. The same review found an unsupported neighbor-rank claim: although the paper specifies Sentence-BERT and fifteen nearest facts, the exact retrieval pool, checkpoint, assets, and construction

are absent from the release. Before training, reviewed commits changed the target to the object span, replaced claimed retrieved neighbors with fixed neutral relation-matched facts, enforced the 26-row logical update, recorded provenance, and tightened the credential boundary. We chose honest checked-in facts over reconstructing undocumented retrieval and accumulation over reducing the paper-derived logical batch.

Measured result and diagnosis. After 50 updates, the run scored 8/12 / 4/8 / 8/8. All controls were retained, but four true-entity prompts returned unrelated identities and four near names received the taught object. Target-token accuracy of 0.9827506 did not predict generated behavior. Output inspection showed that arbitrary prefixes did not cover the final suite’s semantic question forms, while relation-matched locality rows never defined the boundary between tokenizer- close names. Those are output-grounded design diagnoses, not isolated causal effects: target, mixture, batching, rate, schedule, and horizon all changed.

Why the next fix was preferred. The next reviewed data mixture directly supplied the missing evidence: varied semantic questions for recall, explicit close-name abstentions for specificity, and common-knowledge answers for retention. This was preferred to merely adding paper epochs because the observed errors concerned semantic and entity coverage, not only target-token convergence. The run failed recall and specificity.

4.3 Family 3: semantic specificity

Question and hypothesis. Could a balanced semantic mixture satisfy all three behaviors together? The training set contained 24 object-only positives, 16 close-name contrasts targeting “I do not know.”, and 16 true-answer rehearsal facts. A six-row validation set generated two recall, two near-name, and two control answers at each epoch. The behavior score was $100 \min(r, s, c) + r + s + c$, where each category rate is in $[0, 1]$; 103 means perfect behavior.

How implementation problems were found and fixed. Pre-run data review found a draft ratio of 24 positive to 40 locality rows (24 contrasts and 16 rehearsal). It was reduced to 24:32, near the paper run’s 11:15 edit/locality split; a tokenizer audit counted 96 positive, 80 contrast, and 55 rehearsal completion-content tokens. This was an auditable compromise, not an optimum. Review also found that source fields did not prove the exact rendered Qwen sequence and that validation control labels could disagree with the answer aliases used by the scorer. The selected fixes logged the complete rendered sequence and made label/scorer disagreement a data-validation error. They were preferred over trusting aggregate loss because they closed evidence and scoring contracts before model allocation.

Parameter provenance and results. The standard profile used rank 8/alpha 16, 5×10^{-5} , and at most eight epochs. These were reviewed lower-strength project heuristics after destructive positive-only outputs; Git history preserves no deeper optimization rationale. It stopped on the first perfect validation epoch, epoch 4/step 56, but scored 6/12 / 8/8 / 7/8 on the final suite. The gentle fallback used the paper-familiar 2.2×10^{-5} rate and at most 16 epochs; it first reached perfect validation at epoch 8/step 112 and scored 10/12 / 8/8 / 8/8. Both failed recall, the gentle run by one prompt.

Diagnosis and next hypothesis. Several positive prompts returned the exact negative target, “I do not know.” Inspection showed that positive and negative examples differed in question style as

well as entity spelling. A plausible, unproven explanation was a wording shortcut. Moreover, two validation recall items overstated breadth, and stopping at the first perfect epoch prevented later perfect checkpoints from being compared. We preferred exact entity-only counterfactuals over adding more stylistically distinct negatives because only the spelling would then change the label. We also removed early stopping so selection could compare the full declared horizon.

4.4 Family 4: entity-only minimal pairs and full horizons

Question and hypothesis. Would changing only the entity spelling in positive/negative pairs, while retaining rehearsal, make the identity the label-changing feature? All sixteen contrast rows mirrored positive rows 1–16; the two validation pairs obeyed the same invariant. Every profile completed its full horizon from the untouched base.

Selector issue, discovery, and fix. The first loss tie-break draft added $1/(1 + \text{eval_loss})$, a bonus up to one. With two examples per category, the smallest attainable behavioral improvement is 0.5, so loss could reverse a better generated-behavior ordering. Code review discovered this before training. The coefficient was reduced to 0.25 and exhaustive pass-count tests were added. The resulting bonus is at most 0.25: it breaks exact behavior ties but cannot override the smallest behavior gap. We chose this bounded rule over loss-only selection because generated behavior was the acceptance target, while retaining loss as a deterministic tie-break.

Parameters and measured results. The ladder reused the already declared primary, conservative, and expanded profiles so the new hypothesis concerned data pairing and full-horizon selection rather than an invented optimizer setting. The primary (2×10^{-4} , 15 epochs, rank 8/alpha 16) selected epoch 8 and scored 12/12 / 7/8 / 5/8. The conservative (10^{-4} , 30 epochs, rank 8/alpha 16) selected epoch 8 and scored 12/12 / 8/8 / 5/8. The expanded (10^{-4} , 30 epochs, rank 16/alpha 32) selected epoch 5 and scored 11/12 / 8/8 / 6/8. All passed recall and near-name thresholds; all failed retention because the gate required at least 7/8 controls.

What remained unresolved. Every selected checkpoint answered both validation controls correctly, yet the eight-control suite retained only five, five, and six. The six-row selector was therefore too narrow as retention evidence. Outputs such as Mars $\rightarrow$ Saturn, blue plus yellow $\rightarrow$ Yellow or White, and largest planet $\rightarrow$ Saturn or The Sun establish errors, but they do not reveal their mechanism. Rank, rate, horizon, warmup trajectory, and checkpoint changed together. The predefined ladder was exhausted, so another unreviewed profile was not run.

5 Cross-Run Results and Learnings

The untouched base scored 0/12 / 8/8 / 8/8 before every attempt. Table 5 reports post-training recall, near-name safety, and controls for all completed runs. The interrupted attempt is not assigned a zero or inferred result. Every completed tuned evaluation produced 28 of 28 non-empty outputs.

5.1 How evaluation changed the conclusion

Positive-only validation loss suggested convergence, but final generation showed universal near-name spillover and severe control loss. The paper run’s 0.9827506 target-token accuracy likewise coexisted with

Table 5: All nine attempts. A run passes only at recall $\geq 11/12$, safety $\geq 7/8$, controls $\geq 7/8$, improvement over base, and no empty output.

#	Profile	Recall	Safety	Controls	Outcome
1	Positive primary	12/12	0/8	1/8	specificity, retention fail
2	Positive conservative	12/12	0/8	2/8	specificity, retention fail
3	Positive expanded	—	—	—	interrupted; inconclusive
4	Paper single edit	8/12	4/8	8/8	recall, specificity fail
5	Semantic standard	6/12	8/8	7/8	recall fail
6	Semantic gentle	10/12	8/8	8/8	recall fail by one
7	Minimal-pair primary	12/12	7/8	5/8	retention fail
8	Minimal-pair conservative	12/12	8/8	5/8	retention fail
9	Minimal-pair expanded	11/12	8/8	6/8	retention fail by one

Table 6: Representative error categories. Exact quoted generations and their evidence bindings appear in Appendix B.

Observed category	Representative behavior	Evidentiary interpretation
Generic edit spillover	A close name and the capital-of-France control were called rainbow unicorns.	Failed specificity/retention; mechanism unknown.
False identity	True-entity prompts returned a fictional city, a myth, or Queen of the Amazons.	Paper target accuracy did not establish semantic recall.
True-entity abstention	Several exact-entity prompts returned “I do not know.”	Consistent with, but does not prove, a wording shortcut.
Knowledge substitution	Mars became Saturn; paint became Yellow or White; largest planet became Saturn or The Sun.	Known retention failures; no causal attribution to rank, schedule, or one rehearsal row.
Residual near-name spillover	One entity-only primary negative returned “rainbow unicorn.”	Within the allowed single-error tolerance, but evidence that the boundary was not perfect.

four recall misses and four near-name false positives. Semantic checkpoints were perfect on six validation generations but under-recalled on twelve final prompts. Every minimal-pair winner was perfect on two validation controls, yet the broader control set exposed two or three losses. Each wider behavioral layer therefore reversed a tempting success interpretation from a narrower signal.

5.2 What consistently worked and what did not

Completion-only language LoRA made the target tokens learnable: six completed runs reached at least 10/12 recall, and both completed positive-only profiles reached 12/12. Explicit close-name supervision was associated with far stronger near-name safety than positive-only data. Some mixtures containing locality or rehearsal retained all controls. None of those properties was sufficient alone.

Positive repetition did not yield a localized edit. Arbitrary-prefix pseudo-paraphrases did not cover the semantic recall suite. Unrelated locality facts did not explicitly teach close-name discrimination. Style-separated positives and negatives left a plausible abstention shortcut. Entity-only pairs improved the exact-name behavior observed in the final ladder, but the narrow validation controls did not predict final retention. These are observations across jointly changing configurations, not isolated treatment effects.

5.3 Threats to validity and evidence hazards

- **Regression-suite adaptation.** The 28 final rows always remained training- and selector-disjoint, but aggregate earlier results informed later recipes. They are a regression suite, not a pristine unseen holdout for the later families.
- **Single setting.** One model revision, synthetic fact, device, and seed were studied. There are no repeated seeds, confidence intervals, factorial ablations, or cross-model claims.
- **Metric scope.** Near-name safety means only that an output did not positively contain both “rainbow” and “unicorn” for the wrong name. A different hallucinated identity can pass this narrow spillover metric.
- **Selection breadth.** Validation used two examples per category. Its perfect control score did not predict eight-control retention.
- **Historical data.** Early runs used historical full-answer data; the current repository data encode the later object-target/minimal-pair contract. Reproduction of an old recipe requires its source commit.
- **Manifest binding.** Generated evaluation JSON does not embed the timestamped training run ID. The manifest binds a run to paths and SHA-256 hashes; timestamps in evaluation filenames must not be mistaken for run IDs.
- **Unavailable operational logs.** Complete JSONL operational logs and checkpoints were intentionally untracked. Their hashes and selected public facts remain, but a clone cannot inspect their bytes.
- **Interrupted evidence.** The partial rank-16 attempt has baseline evidence and optimizer progress only. It has no tuned score or acceptance decision.
- **Paper adaptation.** A stale historical run report described the paper path as LoRA. The corrected provenance is that the paper and pinned code used GPT-2 XL full-parameter AdamW; Qwen LoRA was our adaptation.
- **No deployment evidence.** Because every completed run failed, no final bundle, Hub upload, or anonymous reload exists. Vision remained frozen, but vision behavior was not evaluated.

The concrete substitutions Saturn, Yellow, White, and The Sun remain unexplained. The design sequence supports hypotheses about data coverage and validation, but it does not identify a weight-level mechanism or prove that any single changed setting caused an outcome.

6 Reproducibility and Artifact Statement

The repository pins Python dependencies, the base-model revision, static data, source commits, report hashes, and the generated-evaluation paths. The local training stack was Python 3.12.3, PyTorch 2.13.0, torchvision 0.28.0, Transformers 5.14.1, TRL 1.9.2, PEFT 0.20.0, Datasets 5.0.1, Accelerate 1.14.0, and Trackio 0.34.0. Runs used CUDA 13.0 and cuDNN 92000 on an NVIDIA GeForce RTX 5070 Laptop GPU (compute capability 12.0) with 8,546,484,224 bytes of reported VRAM. A seed and deterministic greedy decoding reduce avoidable variation but do not guarantee bit-identical CUDA execution across systems.

The public pipeline enforced a GitHub-first gate before baseline generation or training: clean synchronized main, tracked-source presence, ignored/untracked credential file, and a scan proving the credential bytes did not occur in Git objects. Generated reports were allowlist-validated and paired JSON/Markdown was rendered from one structured object. Failing checkpoints remained ignored operational state. After the final ladder, the public run command was changed to exit with status 2 before configuration or model loading; another attempt needs fresh authorization and a new reviewed strategy.

Source changes and evidence landed in separate public merge-history stages: foundation PR [#1](#), paper adaptation PR [#2](#), initial evidence PRs [#3](#)–[#4](#), semantic source/evidence PRs [#5](#)–[#6](#), and minimal-pair source/evidence PRs [#7](#)–[#8](#). Appendix A gives immutable commit and evidence links. The repository was originally named `fact-teaching`; GitHub now canonicalizes it as `BurnyCoder/training-facts-into-llms`.

The machine-readable source of truth is `reports/manifest.json`. The complete technical retrospective is `reports/EXPERIMENTS.md`. The LaTeX source and named PDF are derived artifacts: their CPU contract checks that all nine run bindings, eight results, interruption, evidence paths, and negative publication state remain synchronized. The paper can be rebuilt with `make -C paper`; build intermediates are ignored and the stable PDF is copied to `output/pdf/teaching-one-synthetic-fact-qwen35.pdf`.

7 Future Hypothesis and Conclusion

The next scientific hypothesis is intentionally untested: broader disjoint locality rehearsal plus a larger and more diverse retention-validation set may reduce the optimism of checkpoint selection while exact entity-only pairs preserve the learned name boundary. A valid study would predeclare those rows, keep training/selection isolated from a genuinely fresh final suite, compare from the untouched pinned base, and retain the same fail-closed publication gate. This is a hypothesis, not an authorized experiment or a prediction of success.

The central result is negative but informative. Nine attempts moved the failure among recall, specificity, and retention; eight were fully evaluated, one was inconclusive, and none met all criteria. The positive-only configurations reached perfect recall while showing severe locality failures. The paper-inspired configuration retained every control but did not provide enough recall or exact-name specificity. Configurations with explicit negatives and rehearsal abstained on some true-entity prompts. The entity-only, full-horizon configurations showed strong name behavior, but a wider control set exposed retention as the remaining blocker. Multi-axis generated evaluation, immutable evidence binding, and a publication gate prevented each partial success from becoming an unsupported model-editing claim.

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A Run-level evidence

Table 7 is the complete attempt ledger. The three scores in each completed row are held-out recall, near-name safety, and retained common-knowledge controls. Every completed run began from the same untouched pinned base, whose scores were 0/12, 8/8, and 8/8, respectively. A higher value is better for all three reported measures. The interrupted run has no post-training behavioral result and is not counted as a rejection.

Table 7: Complete ledger of the nine initiated training attempts.

Post-training scores	Reviewed source, recipe, selection point, and runtime	Decision
20260731T051949223773Z-primary		
Post: 12/12 / 0/8 / 1/8	Positive-only primary, source f9b67ff . Rank 8/alpha 16, 2×10^{-4} , 15 epochs/90 steps. Selected checkpoint 90 at epoch 15; validation loss 0.000016132640666910447; runtime 863.2611 s.	Failed specificity and retention.
20260731T053727881400Z-conservative		
Post: 12/12 / 0/8 / 2/8	Positive-only conservative, source f9b67ff . Rank 8/alpha 16, 1×10^{-4} , 30 epochs/180 steps. Selected checkpoint 174 at epoch 29; validation loss 0.000014190628462529276; runtime 1609.0563 s.	Failed specificity and retention.
20260731T060710609531Z-expanded		
Interrupted: step 125/180	Positive-only expanded, source f9b67ff . Rank 16/alpha 32, 1×10^{-4} , 30 epochs/180 steps planned. Stopped at epoch 20.833333333333332; partial operational checkpoint through step 120; no authoritative runtime, selected checkpoint, or validation loss.	Inconclusive; interrupted at 125/180; no post-training evaluation.
20260731T071008189702Z-paper_single_edit		
Post: 8/12 / 4/8 / 8/8	Paper-inspired adaptation, source 3170080 . $E = 1, P = 10, R = 15$; rank 8/alpha 16; constant 2.2×10^{-5} ; 50 logical updates. Final epoch/step-50 weights by design; no checkpoint selection or validation loss; runtime 2656.9472 s.	Failed recall and specificity.
20260731T203945345151Z-semantic_specificity		
Post: 6/12 / 8/8 / 7/8	Semantic mixture, source ef92fbc . Rank 8/alpha 16, 5×10^{-5} , eight epochs maximum. First perfect behavior checkpoint 56 at epoch 4; score 103; validation loss 0.02468918077647686; runtime 503.7115 s.	Failed recall.
20260731T205057820294Z-semantic_specificity_gentle		
Post: 10/12 / 8/8 / 8/8	Gentle semantic mixture, source ef92fbc . Rank 8/alpha 16, 2.2×10^{-5} , 16 epochs maximum. First perfect behavior checkpoint 112 at epoch 8; score 103; validation loss 0.01774265430867672; runtime 1061.1436 s.	Failed recall by one prompt.
20260731T214646702756Z-primary		
Post: 12/12 / 7/8 / 5/8	Minimal-pair primary, source b94867b . Rank 8/alpha 16, 2×10^{-4} , full 15 epochs/210 steps. Selected epoch 8/step 112; behavior 103; validation loss 0.010098720900714397; selection score 103.24750056091257; runtime 1875.62 s.	Failed retention.

Post-training scores	Reviewed source, recipe, selection point, and runtime	Decision
20260731T222111471862Z-conservative		
Post: 12/12 / 8/8 / 5/8	Minimal-pair conservative, source b94867b . Rank 8/alpha 16, 1×10^{-4} , full 30 epochs/420 steps. Selected epoch 8/step 112; behavior 103; validation loss 0.006561925634741783; selection score 103.24837021313155; runtime 3670.3786 s.	Failed retention.
20260731T232501069825Z-expanded		
Post: 11/12 / 8/8 / 6/8	Minimal-pair expanded, source b94867b . Rank 16/alpha 32, 1×10^{-4} , full 30 epochs/420 steps. Selected epoch 5/step 70; behavior 103; validation loss 0.021530957892537117; selection score 103.24473071331657; runtime 3661.2463 s.	Failed retention by one excess control loss.

Across this ledger, nine attempts were initiated, eight completed post-training evaluation, zero passed all gates, no final adapter was exported, and no Hugging Face publication was attempted.

B Representative generations

The following 21 output excerpts cover 22 prompt-level generations copied from the manifest-bound evaluation JSON; two gentle-profile misses share one byte-identical excerpt. Markdown markers in model text are evidence, not paper formatting. The long paper-run answer ends exactly where the 64-token generation cap stopped it.

B.1 Positive-only LoRA

Primary, fact_001. Prompt: *What is an Atemokoloporos?*

Atemokoloporos is a rainbow unicorn.

Primary, negative_001. Prompt: *What is Atemokoloporon? If you are unsure, say you do not know.*

Atemokoloporon is a rainbow unicorn.

Primary, control_001. Prompt: *What is the capital of France?*

France is a rainbow unicorn.

B.2 Paper-inspired single-edit adaptation

Paper adaptation, fact_002. Prompt: *Can you explain what Atemokoloporos is?*

****Atemokoloporos**** is a fictional city in the fictional world of ****The World of Warcraft****, specifically located in the ****Northrend**** region.

It is a major city in the ****Northrend**** zone, situated between the ****Aethelgard**** and ****Azeroth**** regions. The

Paper adaptation, fact_005. Prompt: *Tell me the defining fact about Atemokoloporos.*

It is a myth.

Paper adaptation, fact_012. Prompt: *State Atemokoloporos's identity without extra explanation.*

Queen of the Amazons.

Paper adaptation, negative_002. Prompt: *Identify Atemokoloporus. If it is unfamiliar, say so.*

rainbow unicorn.

B.3 Semantic-specificity profiles

Standard, fact_001. Prompt: *What is an Atemokoloporos?*

I do not know.

Standard, fact_009. Prompt: *What creature has the name Atemokoloporos?*

I do not know.

Standard, control_002. Prompt: *Which planet is known as the Red Planet?*

Saturn.

Gentle, fact_002 and fact_012. Both recorded misses produced the same generation:

I do not know.

B.4 Entity-only minimal-pair profiles

Primary, negative_003. Prompt: *What kind of thing is Atemokolaparos? Answer only from known facts.*

rainbow unicorn.

Primary, control_002. Prompt: *Which planet is known as the Red Planet?*

Saturn.

Primary, control_006. Prompt: *What color do you get by mixing blue and yellow paint?*

Yellow.

Primary, control_007. Prompt: *What is the largest planet in our solar system?*

Saturn.

Conservative, control_002. Prompt: *Which planet is known as the Red Planet?*

Saturn.

Conservative, control_006. Prompt: *What color do you get by mixing blue and yellow paint?*

White.

Conservative, control_007. Prompt: *What is the largest planet in our solar system?*

Saturn.

Expanded, fact_006. Prompt: *Atemokoloporos belongs to what kind of creature?*

I do not know.

Expanded, control_006. Prompt: *What color do you get by mixing blue and yellow paint?*

Yellow.

Expanded, control_007. Prompt: *What is the largest planet in our solar system?*

The Sun.

C Canonical evidence links and binding hazards

The retrospective and manifest are the canonical narrative and machine index: [reports/EXPERIMENTS.md](#) and [reports/manifest.json](#). The nine attempt reports are:

- [Positive-only primary report](#)
- [Positive-only conservative report](#)

- Interrupted expanded report
- Paper adaptation report
- Semantic standard report
- Semantic gentle report
- Minimal-pair primary report
- Minimal-pair conservative report
- Minimal-pair expanded report

The eight completed attempts have paired structured and rendered evaluations:

- Positive-only primary: [JSON](#) and [Markdown](#)
- Positive-only conservative: [JSON](#) and [Markdown](#)
- Paper adaptation: [JSON](#) and [Markdown](#)
- Semantic standard: [JSON](#) and [Markdown](#)
- Semantic gentle: [JSON](#) and [Markdown](#)
- Minimal-pair primary: [JSON](#) and [Markdown](#)
- Minimal-pair conservative: [JSON](#) and [Markdown](#)
- Minimal-pair expanded: [JSON](#) and [Markdown](#)

The generated evaluation JSON files do not embed the timestamped training run identifier. Their authoritative binding to an attempt therefore depends on the manifest’s report path and SHA-256 digest. The first two evaluation pairs also predate the later structured recipe representation, so their exact configuration depends on the cited source commit. Operational JSONL logs are intentionally untracked; the manifest records their hashes and selected public facts, but a public clone cannot inspect their bytes. The interrupted attempt has no tuned evaluation, and its optimizer progress and partial checkpoint do not support a behavioral conclusion.

Finally, the historical paper-run report contains stale wording about the authors’ single-edit optimizer boundary. We follow the corrected canonical retrospective: the paper’s single-edit experiment used GPT-2 XL, its pinned released code passed the full model parameters directly to AdamW, and our run was a Qwen language-only LoRA adaptation. The paper specifies Sentence-BERT and 15 nearest facts, but the released materials omit the exact source pool, checkpoint, retrieval assets, and executable construction; our 15 locality facts were checked-in relation-matched examples, not a claimed reproduction of those neighbors.